

APPLICATIONS ANALYSIS OF PLANT ABUNDANCES OF A SAVANNA WOODLAND IN ZIMBABWE USING GENERALIZED LINEAR MODELS

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Key words: Abundance; generalized linear model; herbivores; link function; multi-species; Poisson distribution; woody.

Summary: Seven fence-lines separating areas with different large herbivores were selected from around Zimbabwe. The numbers of each woody plant species were recorded in plots placed on both sides of each fence-line at 18 randomly chosen positions for each of the seven fence-lines. It has been verified that a Poisson distribution with parameters depending on soil and stocking rate covariates can be used for modelling the number of plants of each species. A generalized linear model was used for the evaluation of the influence of large herbivores on the number of plants. The model works well and highlights the effects of herbivore species, relative stocking rates and overall stocking rate in determining the numbers of plants.

1. Introduction

There has been growing global concern about the rising extinction of organisms. Due to the increasing human population, systems that were

STMA (2004) SUBJECT CLASSIFICATION: 07:095(14:060)

previously relatively untouched are now subject to a variety of management forms. The World Wide Fund for Nature (WWF) set up the Multispecies Animal Production Systems Project (MAPS) to examine how systems of large animal production can be used as ecologically and economically sustainable land-use options (Cumming 1988). The data for this paper were collected under this project to examine the hypothesis 'Systems with diverse assemblages of large herbivore species have a greater species richness of woody plants than systems with a single large herbivore species.' Data were collected at five fence-lines separating game and cattle areas and at two fence-lines separating herbivore communities with different stocking rates and mixtures of large herbivores. The objective was to examine the effects of large herbivores on woody plant diversity. This gave a total of seven study areas that will be referred to as "sites" in the rest of this paper. The data were collected from sets of paired plots. That is, plots were placed at the same position on both sides of the fence separating the different adjacent herbivore systems. Figure 1 shows *Brachystegia spiciformis* in two adjacent plots at McIlwaine National Park (Figure 1a) and Idaho Farm (Figure 1b).

Richardson-Kageler (1999) reported that the woody species richness of these sets of paired plots did not differ although there were differences in the abundances of some of the woody species. The analysis was focused on testing whether there were differences in plant species diversity (species richness and relative abundances). Abundances were examined through the use of a dominance index (the number of plants of a species divided by the total number of plants of all species). The reasons for the differences in abundances of individual woody species was not established and no covariates were examined.

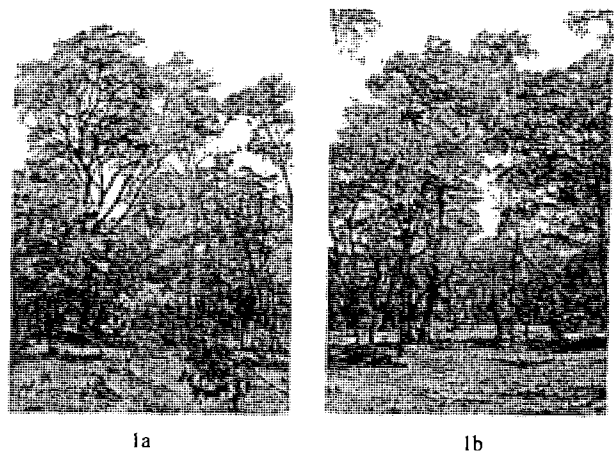


Figure 1. Photographs showing two plots adjacent to the fence-line at Mellwaine National Park (1a) and Idaho Farm (1b)

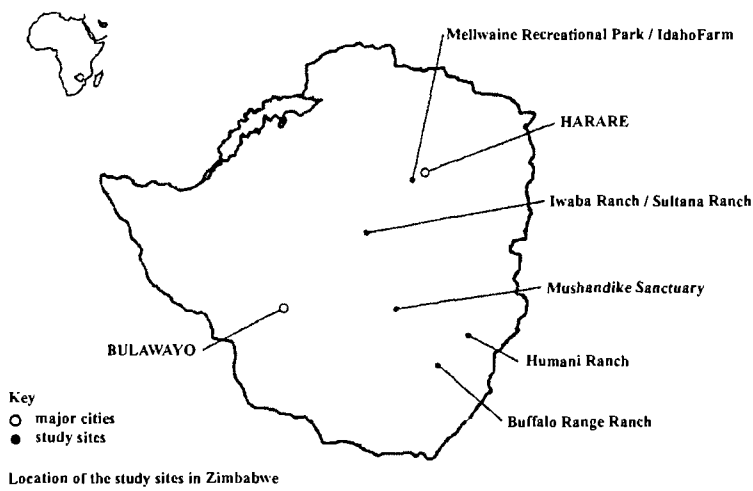


Figure 2. Location of the study sites in Zimbabwe

Magurran (1988), Hill and Hamer (1998) found that changes in species abundances indicated environmental disturbances. Kempton (1979) suggested that relative abundances of species rather than the loss of a species is a more sensitive measure of this disturbance. It is therefore important that the abundances of individual species be examined to establish why they are changing if, indeed, they change. Possible reasons for the changes in woody plant abundances include the stocking rate, the herbivore systems and various soil variables. Thus the main goal of this paper is to investigate the influence of the stocking rate and the herbivore system on the number of individual plant species.

The data on woody plant abundances are typical of ecological data in that there is a great degree of variability from one plot to another within the same herbivore system and there are a large number of zeros in the data matrix. Consequently use of the classical parametric tests based on the normal distribution is not effective. Woody plant abundance is a count measure per unit area for which the Poisson distribution is a natural choice. Thus, the appropriate technique for these data is a Poisson regression. Woody plant abundances in each plot were classified by means of cluster analysis into groups according to the covariates (stocking rate, percentage of sand and percentage of clay). Each group was characterized by similar values of the covariates. If the group was large enough it was tested to see if the data for each woody species were from a Poisson distribution. All the results of goodness-of-fit tests for the Poisson distribution are not included here. The most important results which justify the use of the Poisson distribution are discussed in the paragraph 6. Because the Poisson distribution fits most of the data, it was decided to fit a conditioned generalized linear model (GLM) with a

log link function (Poisson regression) to model the number of plants of a given species in a given plot, taking account of the paired plot design.

Table 1. Details of the study sites

Site	Code	Stocking rate [kg ^{0.75} km ⁻²]	Sand %	Clay %
Buffalo Range Ranch (game)	BRRg	1257	43.28	35.83
Buffalo Ranch (cattle)	BRRc	1181	47.72	38.17
Humani Ranch (game)	HRg	922	57.06	30.78
Humani Ranch (cattle)	HRc	772	65.89	28.83
Iwaba Ranch (game)	IR	606	52.83	35.94
Sultana Ranch (game and cattle)	SR	1588	55.50	35.44
McIlwaine National Park (game)	MNP	1211	80.17	14.39
Idaho Farm (cattle)	IF	3157	70.78	21.39
Mushandike Research Area (game)	MRA	1625	52.17	33.17
Mushandike Ranch (cattle)	MR	814	59.28	30.83
Mushandike Research Area (game)	MRA	1625	48.83	29.11
Rabe (cattle)	RA	977	54.17	30.72
Mushandike Research Area (game)	MRA	1625	67.39	23.50
Mushandike Sanctuary (game)	MS	121	70.94	21.28

For the purpose of this model, the herbivore system is indexed by j , $j = 1$ for the game side of the fence and $j = 2$ for the other side of the fence. Five of the sites had cattle on the other side of the fence, one site had both game and cattle there while the seventh site also had game there, but at a lower stocking rate than at the $j = 1$ side. Due to the nature of the data collection, the herbivore system (side of the fence) cannot be distinguished from the stocking rate as both were constant for all the plots on one side of the fence.

The six plant species that were studied were selected because they were present in total numbers of over 30 at more than one site (so that results could be compared across study areas), or because they were dominant plants or nutritional stepping stones. Nutritional stepping stones are vegetation components that animals eat at different times of the year (Owen-Smith and

Cooper 1989). *Acacia nilotica* and *Dichrostachys cinerea* are important for their pods at the end of the dry season when no other food is available. *Brachystegia spiciformis* is a dominant plant in some areas. *Parinari curatellifolia* is a palatable evergreen plant which would form part of the diet of browsers during the dry season. *Burkea africana* and *Terminalia sericea* are relatively unpalatable, although common, plants and they are eaten when other food is scarce. For each of these species, a Poisson regression model with covariates (herbivore system, stocking rate, percentage of sand and percentage of clay) was fit. The model is developed in this paper and is used to predict plant numbers.

2. Study Sites

Five study sites were used. A brief description of the sites is given in Table 1 and the geographical locations of the sites are shown in Figure 2. The herbivore system at each site had been in operation for at least ten years. Each site had one fence-line with different adjacent herbivore systems, except Mushandike which had three fence-lines. The total number of fence-lines sampled was seven, and each fence-line will be referred to as a site.

3. Data Collection

The collection of the data is described in more detail in Richardson-Kageler (1999). The method was adapted from Mueller-Dombois and Ellenburg (1974). Along each side of the seven fence-lines, 18 positions were randomly selected. At each position a 3 m by 20 m plot was laid out on either side of the fence in such a manner as to avoid paths or roads running along the fence. Thus, at each fence-line there was a total of 36 plots, the data from which are arranged in a 36×10 matrix. The first 4 columns of the data matrix are

occupied by values of covariates and the next 6 columns by the responses, i.e. by the number of each of the six woody plant species.

4. Model

The numbers of any given plant species that were recorded at each plot and on each side of the fence of the 7 study areas are considered to be random variables with Poisson distributions. For testing differences between recorded numbers of plants of a given species on both sides of the fence the GLM model with the log link function and with a set of explanatory variables has been constructed. (See McCullagh and Nelder 1989, Dobson 1990.) The study areas will be considered separately and for each study area the six species of plants will also be considered separately. Thus, let Y_{ij} denote the number of plants of a given species recorded at the i th plot, $i = 1, \dots, n$ ($= 18$), on the j th side of the fence ($j = 1$ for the game side of the fence and $j = 2$ for the other side of the fence). The random variables Y_{ij} are assumed to be independent, each with a Poisson distribution $Po(\lambda_{ij})$ with parameters $\lambda_{ij} = E[Y_{ij}]$. The expectation λ_{ij} of each Y_{ij} can depend on a specific plot i because the position of the plot was chosen so that the conditions for plant growth (such as slope, aspect, altitude) would be similar on the two sides of the fence. However, these conditions can be different for different positions. Thus, only j (the side of the fence on which the plot is situated) influences the herbivore system or stocking rate. Furthermore, λ_{ij} can depend on the set of explanatory variables which reflect different conditions for plant growth on each side of the fence and at each selected place. The vector of explanatory variables recorded at the i th plot on the j th side of the fence is given by the column vector $\mathbf{x}_{ij} = (x_{ij1}, x_{ij2}, x_{ij3})'$, where x_{ij1} , x_{ij2} , x_{ij3} are stocking rate, percentage of sand and percentage of clay at the i th plot on the j th side of

the fence respectively. Further, it will be assumed that the dependence of λ_{ij} on the response variables can be expressed by means of unknown parameters $\beta_1, \beta_2, \beta_3$. Taking all the assumptions together, we assume a GLM with log link function for λ_{ij} as follows:

$$\ln \lambda_{ij} = \alpha_i + \beta' x_{ij} \quad (1)$$

for $i = 1, \dots, n, j = 1, 2$ and a vector $\beta = (\beta_1, \beta_2, \beta_3)'$. Here $\alpha = (\alpha_1, \dots, \alpha_n)'$ is a vector of incidental parameters describing the influence of the plot location i . Among the parameters, the main attention is focused on β_1 . Because the stocking rate x_{ij1} is assumed to be constant on each side of the fence, the parameter β_1 reflects the effect of the differing herbivore systems on the two sides of the fence. Thus the null hypothesis in which we are primarily interested, is $\beta_1 = 0$. Because the different numbers of given species on each side of the fence depends on soil conditions as well, the other hypotheses we need to test are $\beta_2 = 0$ and $\beta_3 = 0$ in order to check if the percentage of sand and clay, respectively, has a significant influence on the number of plants on each side of the fence.

5. Standard Results for MLE and Deviance

The parameters $\beta_1, \beta_2, \beta_3$ were estimated using method of maximum likelihood. For the GLM, it is well known that these estimates have an asymptotic normal distribution. This enabled us to compute asymptotic confidence intervals for the parameters β_1, β_2 and β_3 . The deviance (log-likelihood ratio statistic)

$$D = 2[l(\hat{\beta}_{max}) - l(\hat{\beta})],$$

has an asymptotic χ^2 distribution with appropriate degrees of freedom and it is often used to assess the goodness-of-fit of the model. In general, the χ^2

approximation is a rather crude method for assessing the goodness of fit. (See Dobson, 1990 p. 59 and 60.) That is why other methods are often more useful. Especially, we will examine residuals in cases when the results based on D reject the model. In such cases we will use Pearson and Anscombe residuals. The latter are very useful in the Poisson case because they allow us to make simple residual analyses based on a normal distribution.

Tests for model parameters will be based on an analysis of deviance. See, for example, Dobson (1990) or MacCullagh and Nelder (1989). Let us consider the hypothesis $H_0 : \beta = 0$. Then, if D_0 denotes the deviance (log-likelihood ratio) statistic D under H_0 , the statistic $\Delta D = D_0 - D$ will be used to test if the covariates have a significant influence. In our analysis the statistic ΔD has asymptotically a χ^2 distribution with 3 degrees of freedom.

It follows from the model (1) and from Table 1 where the values of covariates are involved, that the expectations ratio $\frac{\lambda_{i1}}{\lambda_{i2}}$ does not depend on i . Thus the expectations ratio $\frac{\lambda_{i1}}{\lambda_{i2}}$ is a suitable characteristic for comparing the numbers of plants on each side of the fence. This ratio can be expressed as a product $\frac{\lambda_{i1}}{\lambda_{i2}} = M_1 M_2 M_3$ where $M_k = \exp \{ \beta_k (x_{i1k} - x_{i2k}) \}$ represents for $k = 1, 2, 3$ the multiplicative contributions of stocking rate, percentage of sand and percentage of clay respectively to the expectations ratio $\frac{\lambda_{i1}}{\lambda_{i2}}$.

6. Results

The statistical test to verify how good the fit is, was based on the deviance statistic D . The test was performed for each plant and each site. The results are presented in Tables 2–7. The analysis was supported by residual analyses which have been performed for all models. Both Anscombe residuals and Pearson residuals were calculated. The residual plots for the cases where the deviance was significant at the 5% significance level, are presented in

Figure 3. The Poisson distribution was not confirmed for three models, namely *Brachystegia spiciformis* at MRA/MR, *Terminalia sericea* at IR/SR and *Dichrostachys cinerea* at BRRg/BRRc. Goodness-of-fit tests for the Poisson distribution can be performed only if the plant abundance is high and if the covariate values are very similar. Thus the observed discrepancies from a Poisson distribution could be due partly to the differing values of the covariates. It is also necessary to consider the deviation from the Poisson distribution by means of a residual analysis. Based on the Anscombe residuals shown in the Figure 3, it is clear that some of the significant differences were caused by outliers. The most significant Anscombe residuals were found for *Terminalia sericea* at IR/SR and for *Brachystegia spiciformis* at MRA/MR. A similar situation occurs in another four cases where the D statistic was significant at the 1% significance level (see Figure 3). These are *Acacia nilotica* at IR/SR, *Brachystegia spiciformis* at MRA/RA, *Burkea africana* at MNP/IF, and *Terminalia sericea* at MRA/MR. In another three cases (*Acacia nilotica* at MRA/MR, *Terminalia sericea* at MNP/IF and *Dichrostachys cinerea* at MRA/MS) the D statistic was significant at the 5% level. On the other hand, in none of these cases the outliers are excessively large. As a consequence of outliers in the data, a large fluctuation of predicted values is evident (see Figure 4). Thus, these models enable one to predict the numbers of plants, albeit with a high level of uncertainty.

The results for the six plant species will now be described separately. *The * and ** notations are used throughout to denote results that are significant at the 5% and 1% significance level respectively. In all the tables R denotes the stocking rate, S the percentage sand, C the percentage clay and M_• the multiplicative contribution. Sites at which no significant results occurred, are omitted from the tables. Due to space constraints, a plot of observed versus*

predicted numbers is shown only for *Acacia nilotica*. The corresponding plots for the other plant species are obtainable from the second author upon request.

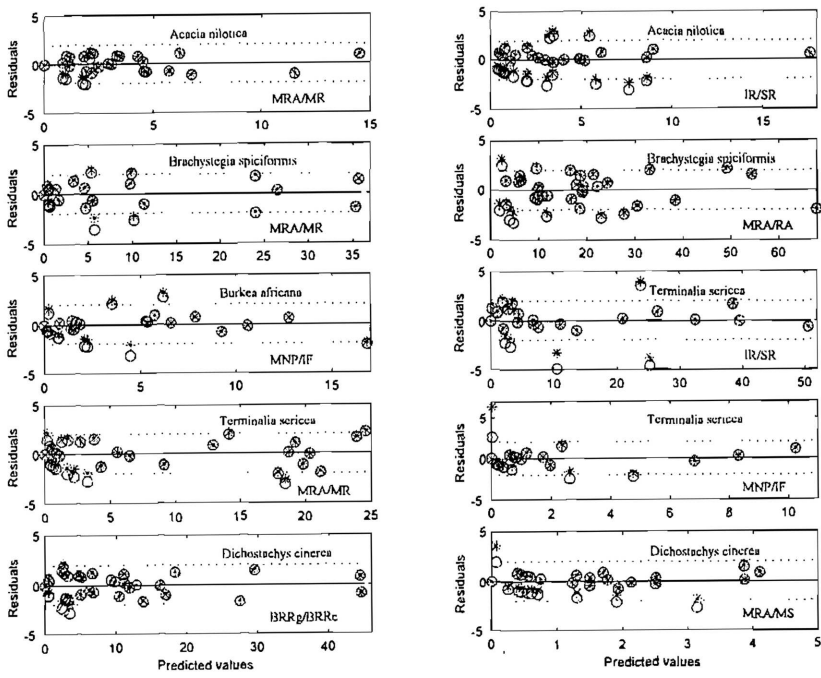


Figure 3. Observed numbers of plants with Pearson residuals* and Anscombe residuals° for models with a significant D-statistic. The dotted lines represent 95% asymptotic confidence intervals based on the Anscombe residuals.

Acacia nilotica was present at four sites and the model predicts well for sites MRA/MS and MRA/MR. The predictions for sites IR/SR and MRA/RA are subject to larger variability (see Figure 4). The estimate of the parameter β_1 was positive at all four sites. The stocking rate covariate was significant only for study area MRA/MS (see Table 2). When the parameter for the stocking rate is positive and the game side of the fence ($j = 1$) has a greater stocking rate than the other side, this would indicate a combined influence of stocking rate and herbivore species which increases plant numbers on the game side compared to the lower stocked side.

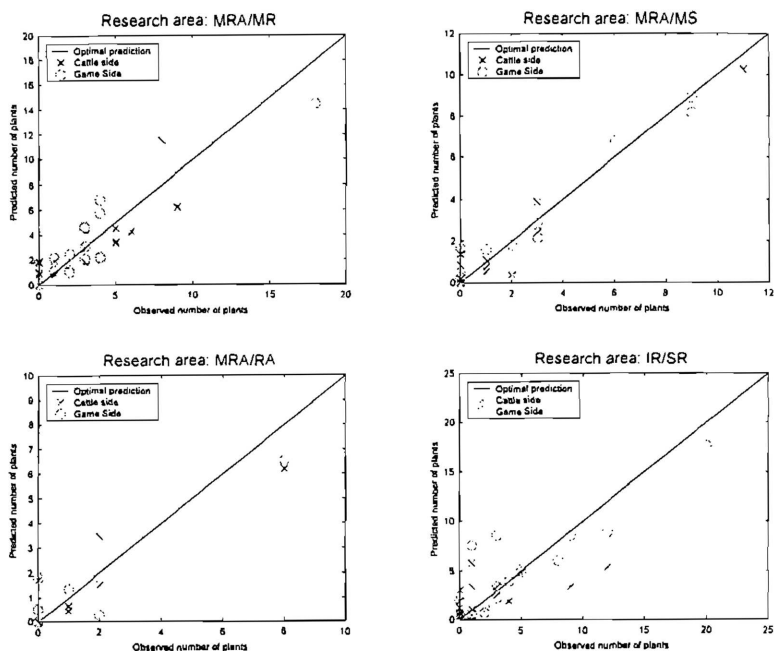


Figure 4. Observed and predicted numbers of *Acacia nilotica*

This species of plant is spinescent and is dispersed through endozoochory (personal observation at Mushandike for eland and buffalo). It is thought that cattle, being ruminants as well, also disperse this species through endozoochory. Thus in higher stocked areas, regardless of whether there were game or cattle present, larger numbers of plants were predicted by the model. The conjecture that the number of plants was significantly influenced by stocking rate and the herbivore system was not confirmed at the other three sites.

Table 2. Results for *Acacia nilotica*

Site	$\Delta D, df = 3$	Covariate	$\hat{\beta}$	$S_{\hat{\beta}}$	$M_{\bullet}$
IR/SR	26.64*	R	0.0003	0.0002	0.74
$D = 68.069^{**}$		S	0.1072*	0.0343	0.75
		C	0.0647	0.0412	1.03
MRA/MR	1.87	R	0.0003	0.0003	1.28
$D = 25.524^{*}$		S	0.0279	0.0283	0.82
		C	0.0328	0.0849	1.08
MRA/RA	5.86	R	0.0007	0.0009	1.57
$D = 16.019$		S	0.0873	0.0494	0.63
		C	0.4580*	0.2322	0.48
MRA/MS	21.05*	R	0.0006*	0.0003	2.47
$D = 16.574$		S	0.0081	0.0918	0.97
		C	-0.1416*	0.0576	0.73

Brachystegia spiciformis is a dominant or co-dominant species in miombo woodlands (Wild and Grandvaux Barbosa, 1967) and was present at five sites. The fit of the model was excellent at MNP/IF, MRA/MS and IR/SR. A higher variability of predicted numbers of plants was evident at sites MRA/RA and MRA/MR. ΔD was significant in all five cases (see Table 3). At the MRA/MS research area, where game are kept on both sides of the fence, larger numbers of the plants were found on the side with the lower stocking rate. Thus, here the increase of stocking rate for game leads to a statistically significant decrease in

plant numbers. A similar situation has been observed at the research area IR/SR. The stocking rate parameter estimate $\hat{\beta}_1$ was significant and the multiplicative contribution was $M_1 = 37.84$, which is very high. This was accompanied by a high influence of sand : $\hat{\beta}_2$ was highly significant and the multiplicative contribution was $M_2 = 0.25$. At the SR side, there was a higher stocking rate than at the IR side and the number of plants was lower. This can be explained by the fact that while cattle are usually classified as grazers, they do increase the amount of browse in their diets at certain times of the year (Fritz *et al.*, 1996). Thus, the plants at SR are browsed by both wild herbivores and cattle.

Table 3. Results for *Brachystegia spiciformis*

Site	ΔD , df=3	Covariate	$\hat{\beta}$	$S_{\hat{\beta}}$	$M_{\bullet}$
IR/SR	70.24*	R	-0.0037*	0.0010	37.84
$D = 21.791$		S	0.5191*	0.1190	0.25
		C	1.0480*	0.1960	1.68
MNP/IF	55.66*	R	-0.0000	0.0004	1.02
$D = 9.778$		S	0.3045*	0.0974	17.45
		C	0.2150*	0.0939	0.22
MRA/MR	15.37*	R	0.0005	0.0003	1.50
$D = 51.559^{**}$		S	0.0625*	0.0188	0.64
		C	0.0519	0.0854	1.13
MRA/RA	50.92*	R	0.0010*	0.0002	1.91
$D = 93.831^{**}$		S	0.0099	0.0060	0.95
		C	0.0006	0.0225	1.00
MRA/MS	32.5402*	R	-0.0006*	0.0003	0.41
$D = 9.760$		S	0.0870	0.0723	0.73
		C	0.2752*	0.0794	1.84

The stocking rate parameter for the MRA/RA was positive and significant at the 5% level, indicating a lower number of herbivore species on the side of the fence with the lower stocking rate (RA with cattle). This suggests that cattle breeding significantly decreases the number of plants. The multiplicative

contribution of stocking rate combined with the herbivore system at this research area is $M_1 = 1.91$ and the ratio of stocking rates is 1625 : 814. This effect can be explained by the fact that in this area there is a herbaceous plant called *Helichrysum kraussii* which appeared to be competing with the *Brachystegia spiciformis* seedlings, thus reducing their numbers. On the MRA side of the fence, game were browsing *Helichrysum kraussii* and this interaction was not taking place. This would cause the effect of herbivore species and stocking rate to be less in MRA and is indicated by the larger numbers of plants recorded in MRA. At two places (MNP/IF and MRA/MR) the estimated stocking rate parameter was not significant.

Burkea africana was present at three sites. The fit was excellent for IR/SR and good for MRA/RA. The D statistic was not significant at the 5% level. At MNP/IF higher variability of predicted values is evident. ΔD was significant at IR/SR and MNP/IF. At MNP/IF the percentage soil and percentage sand were significant. The stocking rate parameter estimate was not significant at any of the three sites. Thus, the influence of cattle breeding on the number of *Burkea africana* has not been confirmed at these sites.

Table 4. Results for *Burkea africana*

Site	ΔD , df=3	Covariate	$\hat{\beta}$	$S_{\hat{\beta}}$	$M_{\bullet}$
IR/SR	46.64*	R	-0.0015*	0.0004	4.36
$D = 9.832$		S	0.1686*	0.0831	0.64
		C	0.7135*	0.2256	1.43
MNP/IF	30.32*	R	0.0005	0.0004	0.38
$D = 42.405^{**}$		S	0.1374*	0.0627	3.63
		C	-0.1154*	0.0546	2.24

The stocking rate parameter β_1 was significant only for IR/SR and it was negative. This indicates that there would be higher plant numbers predicted at the lower stocked IR. The multiplicative effect was $M_1 = 4.3623$. This

means that the big stocking rate in the IR plots with game and cattle has a significant influence on the number of plants. That could be confirmed by Walker (1980) because it is thought that *B. africana* is eaten, but not preferred, by herbivores.

Dichrostachys cinerea was the only species present in all six study areas. The Poisson fit was very good for 4 study areas (MRA/RA, HRg/HRc, MRA/MR and IR/SR) and less so for the other two areas (MRA/MS and BRRg/BRRc). ΔD was significant at IR/SR and MRA/MR. At MRA/MR the stocking rate parameter estimate $\hat{\beta}_1$ was not significant while $\hat{\beta}_2$ and $\hat{\beta}_3$ both were. The multiplicative contributions of sand and clay percentages were 1.74 and 0.55 respectively. The stocking rate at IR (game) was 606 [$kg^{0.75} km^{-2}$] and 1500 [$kg^{0.75} km^{-2}$] at SR (game and cattle). The multiplicative contribution was $M_1 = 0.3078$. Thus it appears that *D. cinerea* occurs in higher numbers in higher stocked areas. *D. cinerea* is a bush encroachment species which increases in areas of overstocking. The pods are a valuable source of food when other food is short (Coates Palgrave, 1988), and the seeds have been collected and germinated from ruminant dung. Higher numbers of plants have been predicted at the side with the higher stocking rate. This has been confirmed at BRRg/BRRc and at MRA/MS but only at the 10% significance level.

Table 5. Results for *Dichrostachys cinerea*

Site	ΔD , df=3	Covariate	$\hat{\beta}$	$S_{\hat{\beta}}$	$M_{\bullet}$
IR/SR	25.00*	R	0.0012*	0.0003	0.31
$D = 17.894$		S	0.0552	0.0632	0.86
		C	-0.0114	0.1202	0.99
MRA/MR	9.36*	R	-0.0005	0.0003	0.67
$D = 10.493$		S	-0.0779*	0.0345	1.74
		C	-0.2578*	0.1119	0.55

Table 6. Results for *Parinari curatellifolia*

Site	ΔD , df=3	Covariate	$\hat{\beta}$	$S_{\hat{\beta}}$	$M_{\bullet}$
MNP/IF	9.33*	R	0.0004	0.0004	0.46
$D = 13.990$		S	0.0925	0.0707	2.38
		C	0.1168	0.0759	0.44
MRA/MS	25.66*	R	0.0004	0.0005	1.82
$D = 20.02$		S	0.1260	0.0940	0.64
		C	0.2668*	0.1217	1.81

Table 7. Results for *Terminalia sericea*

Site	ΔD , df=3	Covariate	$\hat{\beta}$	$S_{\hat{\beta}}$	$M_{\bullet}$
IR/SR	44.91**	R	-0.0004*	0.0001	1.48
$D = 82.092^{**}$		S	0.0683*	0.0194	0.83
		C	0.2932*	0.0514	1.16
MNP/IF	20.97*	R	0.0034*	0.0010	0.00
$D = 25.070^{*}$		S	0.5328*	0.1818	148.86
		C	-0.0654	0.1405	1.58
MRA/MR	3.49	R	0.0001	0.0002	1.08
$D = 63.922^{**}$		S	0.0569	0.0388	0.67
		C	0.0433	0.0694	1.11
MRA/RA	15.23*	R	-0.0002	0.0005	0.88
$D = 3.545$		S	-0.0542	0.0328	1.34
		C	0.5052*	0.2280	0.44

Parinari curatellifolia was present at two sites (MNP/IF and MRA/MS) and the models were not rejected by the goodness-of-fit statistic D . ΔD was significant for both of them (Table 6). The stocking rate parameter was not significant at either site. But both values of $\hat{\beta}_1$ indicate that a higher stocking rate brings lower numbers of plants. The models showed relatively good predictions. Only $\hat{\beta}_3$ was significant (positive). Because *Parinari curatellifolia* is a preferred browse species of both cattle and game (personal observation), it could have been expected that the species would be affected by both the herbivore system and the stocking rate. However, it is possible

that seed dispersal occurs through spitting when cattle chew cud (as has been observed in Eland). This could explain why the stocking rate parameter was not significant, because the seed dispersal would tend to cancel out the browsing effect.

Terminalia sericea was present at five sites. The fit for MRA/RA was excellent and that for MRA/MS was good. For MNP/IF, MRA/MR and IR/SR the values of the D statistic were significant. The estimate of β_1 was significant for IR/SR and MNP/IF (see Table 7). Despite the significantly negative stocking rate parameter estimate for IR/SR, the latter showed higher plant numbers at IR which is lower stocked than SR (and which has both cattle and game). This apparent anomaly could be explained in terms of the germination of seedlings. There were 48 plants less than 50 cm tall at MRA and 25 plants at MS and it is known that litter affects the germination of seeds and growth of seedlings and adult plants (Bergelson, 1991). There was very little litter at MRA ($1625 \text{ kg}^{0.75} \text{ km}^{-2}$) compared to that at MS ($121 \text{ kg}^{0.75} \text{ km}^{-2}$). The litter present at MS may inhibit germination of seeds and growth of seedlings and hence there would be less seedlings in the plots in this area which has the lower stocking rate. The litter effects could be cancelling out the browse effects.

7. Conclusion

New results have been presented regarding the effects of stocking rates of herbivore systems on the abundance of some woody plant species. It has been found that the abundance of *Brachystegia spiciformis*, *Burkea africana* and *Terminalia sericea* at the study sites depend on the stocking rate while the abundance of *Parinari curatellifolia*, *Dichrostachys cinerea* and *Acacia nilotica* seem not to. When the study areas were compared the most significant

influence of the stocking rate and herbivore system was found at IR/SR. It was possible to compare the influence of game at IR with that of game and cattle at SR. At this site the abundances of four of the five plant species present were significantly influenced by the stocking rate and the herbivore system. A further model needs to be developed to distinguish between stocking rate and herbivore species as these could not be distinguished for individual sites. Deeper analysis would be possible if repeated measurements in the studied areas were available.

Acknowledgements

The authors would like to thank an associate editor and two referees for many comments that led to a much improved exposition of the work. They also wish to thank the Masaryk University for support in the writing of this paper. The data were collected under the Multispecies Animal Production Systems Project (World Wide Fund for Nature). Mr. M. Mitchell took the photographs in Figure 1.

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